

Quantifying long-term market impact*

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ABSTRACT

Impact costs occur when large buy or sell orders move market prices. The measurement of these costs is crucial for the evaluation of potential trading strategies as well as the successful execution of systematic investment strategies. However, common approaches suffer from a type of myopia: impact is only measured for the current transaction. In many cases, orders are correlated and the impact of the first order will affect the execution of future orders. We propose a new measure that quantifies the long-term effects of market impact: Expected Future Flow Shortfall (EFFF). Our method is both intuitive and straightforward to implement. Importantly, the EFFF method performs competitively with far more complex and data hungry approaches. Our method should be useful for both the evaluation of execution methods as well as the sizing of orders.

Keywords: *Impact costs, slippage, square-root law, implementation shortfall, metaorders, child orders, changepoints, systematic strategies, alpha, trading strategies, optimal execution.*

JEL: *G11, G12, G13, G23, C58*

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Introduction

The easiest way to boost the alpha of a trading strategy is to reduce costs. While it is well known that execution costs are important, there is relatively little research on execution compared to the signal development. Remarkably, in most published papers, execution costs are routinely ignored. With the rise of systematic trading and machine learning driven investment strategies (Lopez de Prado 2018), investors are moving towards faster, higher turnover strategies. For those strategies, the effects of trading costs are even more important and potentially can render some strategies unprofitable if the costs are not properly considered.

We propose a new method to measure long-term impact costs. The fact that one trade can impact the price of the next trade is not new. Modern trading systems routinely break larger orders (metaorders) into sequences of smaller (child) orders. This introduces autocorrelation at the level of the individual child orders (see Almgren and Chriss 2001). However, for systematic strategies even the metaorders can exhibit autocorrelation over longer time horizons and this is what we refer to as *long-term* impact cost. Our method, Expected Future Flow Shortfall (EFFF), is designed to measure these longer-term costs.

Execution costs can be divided into direct costs, which include commissions, fees and taxes, and indirect costs, which include the bid-ask spread and market impact. There is an existing literature on the effects of both direct and indirect execution costs (see Grinold and Kahn 1999, Bouchaud et al. 2018, and references therein). Market impact is simply the disturbance trading has on prices, in other words the difference between prices observed during and after trading and the prices which would have been observed had you not traded. This quantity, which is not directly observable, has been the subject of considerable study and is often characterized by the so-called square-root law, where the market impact is seen to grow like the square root of the order size (Torre and Ferrari 1997, Grinold and Kahn 1999, Almgren et al. 2005, Moro et al. 2009). Besides the empirical evidence, attempts have also been made to explain the economic foundations of the square-root law (Kyle and Obizhaeva 2018). In dollar terms, the market impact costs of any reasonably sized orders are typically an order of magnitude larger than direct execution costs.

A classical tool of investment management to measure trading effects is the implementation shortfall metric (Perold 1988). This shortfall reflects the discrepancy in performance between an ideal “paper” portfolio and the real portfolio. The implementation shortfall is the difference between transacting at realized prices and transacting at the model (or decision) prices, usually taken to be the mid-point prices when investment decisions are made. The shortfall can be decomposed into execution costs and opportunity costs. Execution costs relate to already executed trades while opportunity costs relate to unexecuted transactions, i.e., past trading opportunities that were unrealized. Faster execution generally increases execution costs and lowers opportunity costs.¹

The focus of this paper is to provide an analogue of the implementation shortfall that quantifies how the market impact of orders can influence the decision prices and hence trading costs of *subsequent* metaorders. We illustrate this using a simple week-long scenario with a series of metaorders, although our approach is more suited for shorter horizons. We execute one metaorder per day. In this scenario, an investment decision is made using the decision price, e.g., the opening price. Post execution of a given metaorder, the number of shares traded

¹ To give an example, suppose we wanted to buy 10,000 shares of a stock and spread out the execution over the course of the day to avoid market impact. Over the first hour of trading, we executed 1,000 shares, thus 9,000 remain unexecuted. If over the course of the same hour the price moved from \$100 to \$101, we would have incurred an opportunity cost of \$9,000, i.e., we missed out on a price move of \$1 on the 9,000 shares which we had not yet executed.

in that metaorder is multiplied by the difference between the weighted average execution price and the decision price. This is recorded as implementation shortfall for that day. The next trading day, the process is repeated, and the implementation shortfall is calculated using the new decision price. This approach works consistently if the investment decisions taken on subsequent days are independent. However, this approach fails when orders are correlated. That is, the initial trading decision must account for the potential correlations of orders to get a reasonable estimate of the cost of the trade – and the profitability of the strategy.

In practice, ignoring metaorder correlation can lead to flawed implementations of portfolios as well as biased estimates of the capacity of the underlying strategies. Correlation between metaorders can arise in a variety of situations. Typical examples are momentum and reversion strategies, where order flow has positive and negative autocorrelations respectively. More widely, any systematic strategy that employs any form of smoothing on raw signals creates autocorrelations in the order flow. The approach that we propose suggests a practical way of estimating the additional effect on P&L coming from the metaorder correlations.

Trading costs and market impact

We now set up our model and define the terminology. Consider portfolio managers making investment decisions, expressing those in the form of desired trades, i.e., in shares to buy or sell. The decision is communicated to the execution agent (which can be part of the same firm or an external service). The execution agent trades the desired amount using “elementary” actions such as submitting market orders/limit orders to the exchange, cancelling limit orders, etc. If the desired trade is small – for example if it is equal to 100 shares – the whole size is going to be submitted to the exchange in a single order. In the more typical situation for large investors, submitting the whole quantity is impractical (for example, the desired trade size may exceed the total liquidity available at the exchange at that moment in time). In this case, the desired trade is typically split by the execution agent into smaller suborders and executed incrementally. Those suborders are called child orders. We will call the desired trade that was communicated to the execution agent a metaorder.

When the investment decision is made, the latest available market price (such as the latest mid-point price, or previous closing auction price) is noted and we call this the *decision price*. When the instruction arrives at the execution agent, the latest available market price may be noted again and is called the *arrival price*. We assume that the processes of communication are very fast and efficient, so that the decision and arrival prices coincide, and accordingly these terms are used interchangeably in what follows.

We begin our analysis with a single buy metaorder. The portfolio manager cares about how the weighted average execution price of all the spawned child orders compares to the decision price for the metaorder. In practice, the weighted execution price can be extremely variable as we illustrate in Panel A of Exhibit 1 which shows the evolution of 1,000 simulated price moves during the execution of a buy metaorder starting at a decision price of 100.² For buy orders, the price path moves higher in slightly more than 50% of cases whilst moving lower in the remainder. On average, across many metaorders, the average execution price path is

² We generate the trajectories according to the following formula for the price:

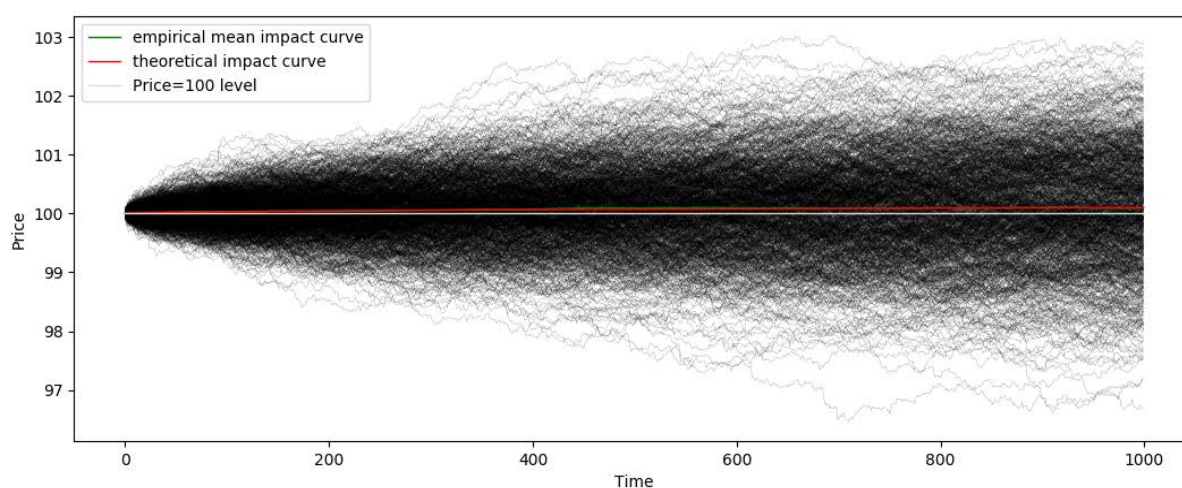
$$P_k = 100 + \sum_{i=0}^{k-1} \sqrt{3/5 \cdot 10^{-3}} X_i + 0.1 \sqrt{i \cdot 10^{-3}}$$

where X_i is the standard student-t distribution with five degrees of freedom. This expression is chosen to show the theoretical price impact at a realistic scale compared to price moves.

higher than the decision price. This is not immediately clear in Panel A because the mean impact curve looks flat but this is simply due to the plotting scale.

In Panel B of Exhibit 1, we focus on a narrower range of prices to see how the impact drives prices higher. Comparing the two panels, it is obvious that the standard deviation of the price moves in Panel A, approximately 100 bps, is much larger than the impact, approximately 10 bps. This illustration is also suggestive of the difficulty we face in measuring price impact curves in practice. An upward change in the asset price that is unrelated to the metaorder may be incorrectly attributed to market impact. Given that the variance of the asset price is so high (compared to the impact cost), attribution to impact is challenging. The difference between the weighted execution price and the decision price expressed relative to the decision price is called *unit slippage* or *unit implementation shortfall* for a buy order. For sell orders, we look at the difference between the decision and weighted execution prices. When multiplied by the quantity traded, this is called the *dollar slippage* or *implementation shortfall*.

Panel A



Panel B

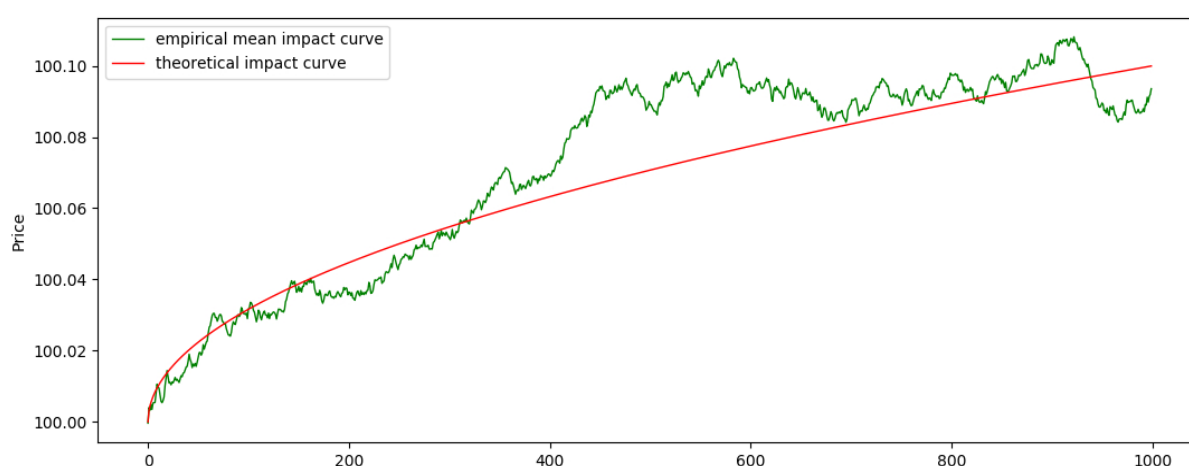


Exhibit 1. Market impact based on 1,000 Monte Carlo simulations. Panel A shows 1,000 simulated mid-price moves (black) during the execution of a buy metaorder. The average of those mid-price paths gives the empirical mean price impact curve that is shown in green, together with the theoretical curve from the square root law in red. These empirical and theoretical impact curves are much clearer in Panel B as that focuses on a narrower range of prices. Note that price impact is around one to two orders of magnitude smaller than typical mid-point price moves, as becomes apparent when comparing the scales of Panels A and B.

There are several contributing factors that make the expected implementation shortfall positive. One is spread cost. For a small metaorder, if an impatient investor submits all the quantity in one market order, the expected unit cost would be half the bid-ask spread. Even for a more patient investor, trying to use limit orders only, the executed price may typically be some fraction of the bid-ask spread. The second cause is market impact. Consider the case of a buy metaorder. The basic laws of supply and demand suggest that greater demand should drive prices up. It is also possible that prices rise more than simple supply and demand considerations would suggest, as other traders – observing prices going up – may infer this to signal future demand (i.e., prices are cheap). Finally, the third cause is alpha. On average, the portfolio manager tries to buy assets when the expectation is for the price to go up.

We primarily focus on the second component – market impact – but consider impact across different metaorders rather than within a metaorder. This is especially relevant for faster trading strategies where one cannot expect that the price impact which results from a single metaorder has decayed away by the end of this metaorder. Instead, the impact is likely to persist to some degree after the end of that metaorder and thereby affects prices for subsequent metaorders.

There is both theoretical and empirical work that examines how market impact evolves during and after metaorder execution. However, there are several hurdles: given the variance of asset prices, a large volume of data on trading decisions and execution results is needed to well-calibrate a model. To make it even more challenging, this information is typically private to an individual investor. If one has only access to child orders of multiple investors without knowing their identity, reconstructing the corresponding metaorders with high fidelity may prove impossible. Without identity, cause and effect can easily be confused.

The broad stylized facts of market impact are as follows (see Bouchaud et al. 2018 for a review): the mid-price is on average increasing throughout the buy metaorder duration and will likely start reverting after the metaorder finishes trading. It may or may not revert completely. The part of the impact that never reverts (or is extremely slow to revert), is called *permanent impact*. The stylized picture for price impact of an isolated buy metaorder is presented in Exhibit 2. The price trajectory over the metaorder can be linear or non-linear (typically concave).

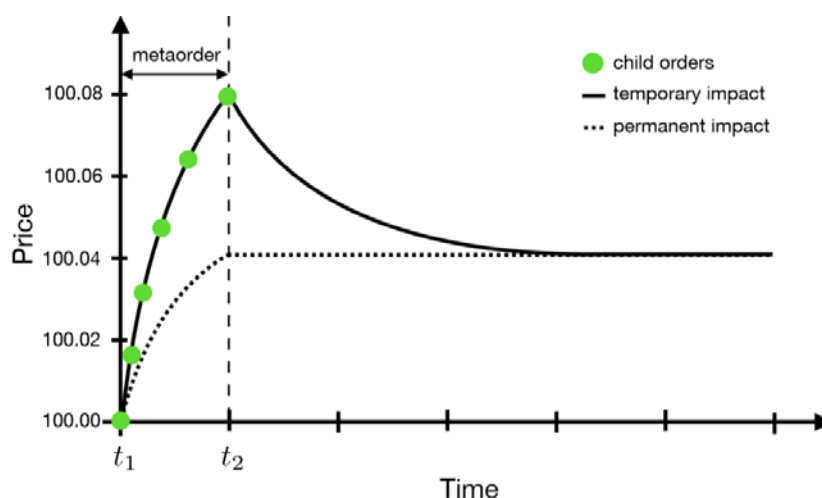


Exhibit 2. The stylized representation of average permanent vs. temporary price impact. The exhibit shows the average effect of the execution of a metaorder consisting of six (child) orders, as well as the resulting average temporary and permanent price impact. The decision price (100) is the price that would be expected at each time point if there was no trading.

Direct measurement of the permanent market impact is practically impossible, as subsequent metaorders may start trading before the reversion of the initial metaorder has finished. For large-scale asset managers, it is often impractical to submit the whole desired amount in one metaorder. Indeed, the amount may not even be possible to trade over a full day. This leads to an autocorrelation effect: buy metaorders tend to be followed, on average, by buy metaorders, and vice-versa for sell metaorders. Exhibit 3 presents a stylized picture for trading two consecutive buy metaorders.

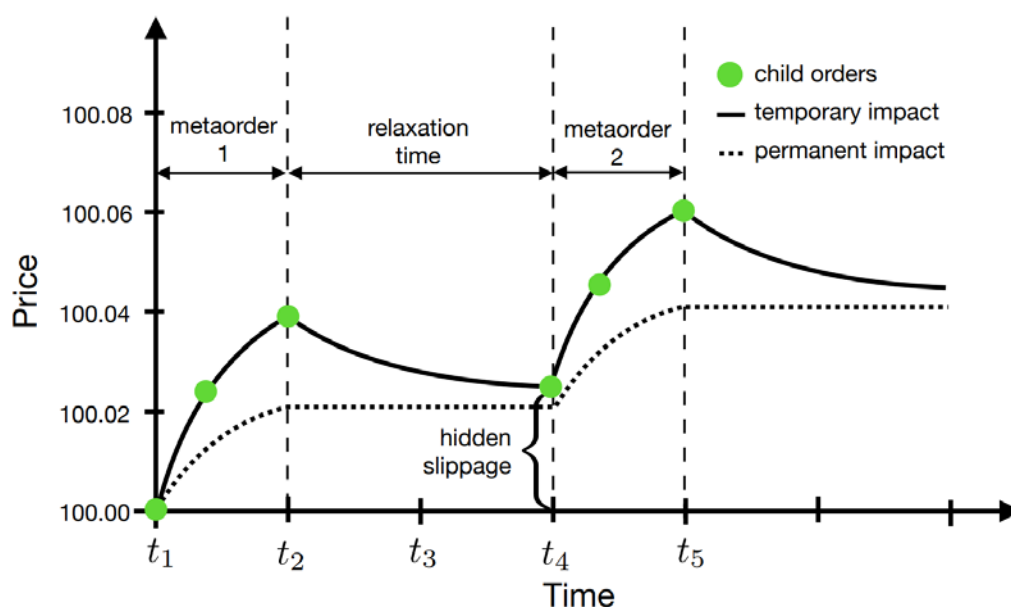


Exhibit 3. Execution of two metaorders with a “relaxation” time in between. The relaxation time, between t_2 and t_4 , is used in practice to allow part of the temporary impact of the first metaorder to decay. However, the decision price of the second metaorder is still affected by the impact of the first metaorder, leading to hidden slippage.

Notice in Exhibit 3 that the decision price of the second metaorder is approximately 100.025 and thus higher than 100 due to the impact of the first metaorder (both temporary and permanent). There are several insights here. First, the difference in price at time t_4 (100.025 to 100) is not reflected in the usual slippage metric for either metaorder. Second, if the profitability of the trade idea depends on the execution of both metaorders, it is essential to consider the impact of the first metaorder on the price used to benchmark the second metaorder. Such considerations may also influence the sizing decision of each metaorder for best overall execution. Research or paper trading often assumes that both metaorders could be executed at a price of 100 plus implementation shortfall. This would misrepresent the viability of the trade idea. This effect on the price, cost and profitability of the subsequent trades is sometimes called *hidden slippage*, as it is not reflected in the traditional slippage metric. The EFFT model enables measurement of this hidden slippage. One might also consider joining the two metaorders into one composite metaorder. Later we discuss the shortcomings of this naïve approach and how the EFFT model overcomes the shortcomings.

EFFT: The Expected Future Flow Shortfall model

Model set-up

We denote the i -th metaorder decision/arrival price by A_i and its size (in shares or lots) by Q_i . Note that we assume sizes to be signed, i.e., to be positive for buy orders and negative for sell orders. We also refer to buy and sell orders as orders of different “side”. Execution costs are measured relative to A_i . We assume that an individual metaorder with arrival price A_i and size

Q_i is broken into a sequence of K_i child orders of sizes $q_{i,k}$ (so that $Q_i = \sum_{k=1}^{K_i} q_{i,k}$) and that these are executed at prices $p_{i,k}$ for each $k = 1, \dots, K_i$ (see Exhibit 4). Note that we assume that metaorders are broken down in child orders of the same side, i.e., if the metaorder is a buy order all the child orders will be buy orders as well. The split of the metaorder into child orders is usually undertaken to reduce total execution costs by smoothing market participation (and thereby impact) across the available execution horizon.

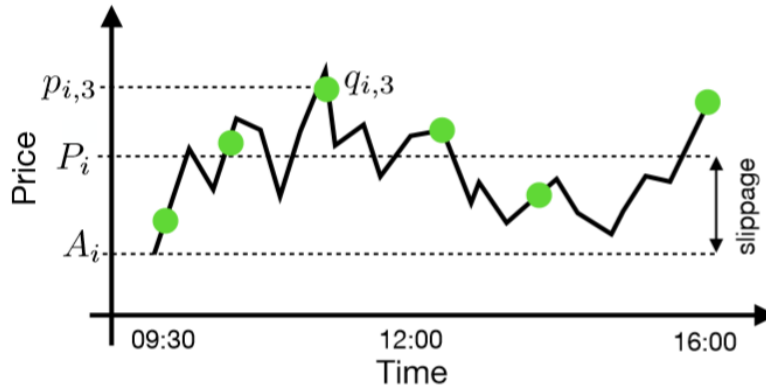


Exhibit 4. Illustration of the execution of the i -th metaorder over time. The arrival/decision price is denoted by A_i . The green circles indicate six buy child orders. For the k -th child order of the i -th metaorder the size and price are denoted by $q_{i,k}$ and $p_{i,k}$ respectively. The volume weighted average fill price of all child orders is denoted by P_i , for example, the third child order has volume $q_{i,3}$ and price $p_{i,3}$.

The volume weighted average fill price of the child orders of the i -th metaorder is therefore given by $P_i = \text{Total Amount Paid} / \text{Total Ordersize} = (\sum_k |q_{i,k}| p_{i,k}) / (\sum_k |q_{i,k}|)$. The dollar slippage or execution cost (ignoring any predetermined fees) of the i -th metaorder is therefore:

$$C_i = (P_i - A_i) \cdot Q_i.$$

Rather than just focusing on the cost of *this* trade, we are interested in the effect of the price move on *future* metaorders (which we call the *expected future flow* or EFF). After execution of the metaorder is completed, the price has potentially moved and the next metaorder starts at arrival price A_{i+1} . The price move $(A_{i+1} - A_i)$ affects subsequent metaorders. We propose the following simple formula to proxy for the effects that this has on the profitability of a strategy:

$$EFF \text{ Shortfall (EFFS)} = \underbrace{(A_{i+1} - A_i)}_{\text{price move}} \times \underbrace{E_i \left[\sum_{j=1}^N Q_{i+j} \right]}_{\text{Expected Future Flow}},$$

where $E_i[\cdot]$ denotes the expected value (at time of A_i) and N is the number of future metaorders to include in the calculation. We typically choose N so that the autocorrelation within the sequence of metaorders at this separation is close to zero, in which case the subsequent contributions to the sum above become small. Here the quantity $E_i[\sum_{j=1}^N Q_{i+j}]$ is the net expected future flow (EFF) at the arrival time of A_i . This measurement can be done *ex post*³

³ Another application of the EFFS formula is where *ex ante* estimates of $A_{i+1} - A_i$ are available. This is, for example, the case when there is an option to choose between two execution algorithms and we know that, on average, they cause different market impact. In this situation, we may use the *ex ante* estimated

once we know the price A_{i+1} . Note that the EFFS for individual trades is very noisy. Because of this, we average it over many trades, say over the course of a month. As a result, computing the EFFS on an ex post basis, day by day, is sufficiently reactive and allows us to use it in real time. This allows us to use the EFFS calibrated on historical data for current trading.

We show how to implement the EFFS formula and how the estimation problem works in practice below. Furthermore, in Appendix A, we present a further simplified version that approximates the EFFS formula with an example application.

Properties of the EFFS formula

The EFFS formula is a natural generalization of the classical implementation shortfall (IS) formula to the case when the trades are not known in advance. Indeed, if the classical IS formula is applied to one composite metaorder consisting of individual metaorders rather than child orders, it can be re-arranged as:

$$\text{composite metaorder IS} = \sum_{i=1}^N (P_i - A_1) Q_i = \sum_{i=1}^N \underbrace{(P_i - A_i) Q_i}_{\text{individual metaorder IS}} + \sum_{i=1}^N \underbrace{(A_{i+1} - A_i)}_{\text{price move}} \times \underbrace{\sum_{j=1}^{N-i} Q_{i+j}}_{\substack{\text{known} \\ \text{Future Flow}}}$$

The first sum in the decomposition (IS of each metaorder) is the slippage to the most recent benchmark price (which includes spread costs). The second sum consists of the terms which structurally differ from the EFFS formula. In the case of IS for the composite metaorder, the quantities Q_{i+j} and N are known. In the context of the EFFS, however, the future flow needs to be estimated, thus leading to the expectation in the EFFS formula. Furthermore, N is also unknown and needs to be chosen or estimated when computing the EFFS. We comment further below on how this relates to alternative approaches such as the stitching approach.

Benchmarking each metaorder to past decision prices

Appendix B provides another perspective on the EFFS formula by the rearranging terms. Each metaorder is decomposed into a sum of many metaorders (potentially of different parity/sign i.e., buy or sell), with each new metaorder having a different decision price. To give a simple example, suppose that on Monday, we expect to buy 1,000 shares on Wednesday. On Tuesday, we revise this to 1200 shares (to be bought on Wednesday). Finally, on Wednesday we end up submitting a metaorder for 1,300 shares. We can interpret this as a sum of three metaorders: first, buy 1000 shares benchmarked to Monday's decision price; second, buy 200 shares benchmarked to Tuesday's decision price; and third, buy 100 shares benchmarked to Wednesday's decision price. If the values in the example instead had been 1,000, 1,300 and 1,200, then the metaorder sequence would have been buy 1,000, buy 300 and sell 100.

Alpha and the EFFS formula

The EFFS will produce a high value if the expected future flow is high and/or the price move between metaorders is large. Here we focus on the price move. A large price move can be caused either by the market impact of trading or by external events which might have been the source of the alpha of the strategy. In the former case, a reduction in the EFFS might be achieved through more efficient execution or through trading in smaller quantities. However, if the EFFS shortfall is high primarily because of the alpha of the strategy, this signifies an effect of opportunity cost and efforts should be focussed on better timing of the investment decisions across metaorders.

quantities $E_i(A_{i+1} - A_i) \sum_{j=1}^{N-i} Q_{i+j}$ combined with *ex ante* slippage estimates for the two algorithms to inform our choice.

Temporary and permanent market impact in the context of the EFFT formula

Estimating the effects on prices long after a metaorder has finished is challenging as the noise around the observed prices increases roughly as a square root of time. Based on several empirical studies (see Bouchaud et al. 2018), it is generally believed that the impact of a trade can be decomposed into two components, consistent with the discussion around Exhibit 2:

- 1) Temporary impact, which eventually decays away; and
- 2) Permanent impact, which does not fully decay. Sometimes the permanent component is theoretically postulated to be a linear function of trade size. However, empirical work suggests the decay behaves like a power law. A number of researchers have tried to model this impact. For example, Farmer et al. (2013) suggest that the permanent impact declines to two thirds of the peak impact.

In practice, impact often declines very slowly over long horizons making it hard to distinguish between temporary and permanent impact. One way to deal with this slow decay is to introduce waiting or relaxation times between metaorders to give time for temporary impact to run its course, as also illustrated in Exhibit 3. However, in practice, it is often impossible to distinguish between truly permanent and slowly decaying temporary impact.

At first glance, one might think that the EFFT is only consistent with a view of the world in which all the market impact is permanent. That is, a permanent price shift arises that impacts all future orders. However, as we demonstrate later in our empirical work, the EFFT formula is also consistent with temporary market impact.

The EFFT formula in context of alternative approaches

Stitching of metaorders

Given the similarity of the EFFT formula to the classic IS formula, why not just apply the IS formula directly to metaorders? An approach popular in industry is to do just that, i.e., stitch together several metaorders into one large composite order based on trade direction (see Gomes and Waelbroeck 2014 and references therein). Exhibit 5 presents an example: a sequence of several buy metaorders in a row (such as denoted by the first four green circles in Exhibit 5) would be redefined as one composite metaorder, with the corresponding decision price defined as the arrival price of the first metaorder in the sequence. Then the standard IS formula would be applied to this new composite metaorder.

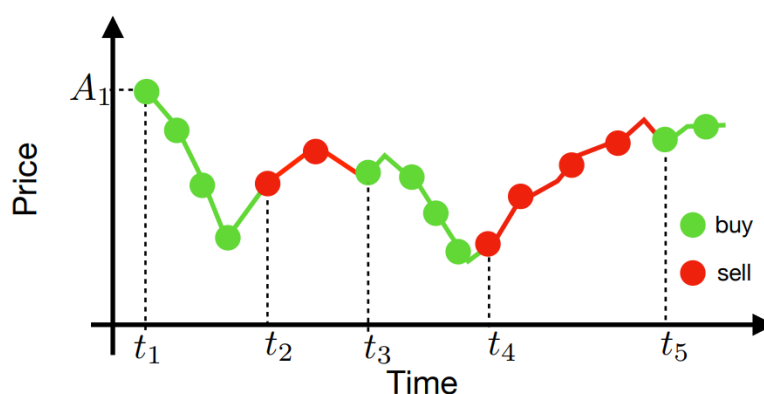


Exhibit 5. Illustration of stitching of 16 metaorders into five composite metaorders. Each green/red circle represents a buy/sell metaorder. The green/red lines connecting buy/sell metaorders represent composite *stitched*

metaorders. This involves grouping together sub-sequences of metaorders that are in the same trade direction. The times $t_1, \dots, t_5$ denote the commencement of each composite order. This example illustrates a simple reversion strategy. Given that on average we sell in rising markets and buy in falling markets, the IS of the five composite metaorders is negative.

Unfortunately, this approach is problematic in several situations. Firstly, suppose that trades are correlated with price moves such as in a mean reversion strategy. In this case, prices may be drifting down over multiple metaorders leading to a longer sequence of buys, followed by sequences of sells when the price is moving up as illustrated in Exhibit 5. This would likely lead to low, or even negative, implementation shortfall when benchmarked to the starting metaorder price. Importantly, this shortfall may have nothing to do with market impact. Second, even for the case of independent metaorders, the IS formula applied to composite metaorders yields an extra bias on P&L, in contrast to the EFFS. Indeed, for completely independent metaorders the IS without any stitching would be the correct measure, as there is no autocorrelation. However, if, purely by chance, we observe consecutive orders of the same direction that are then stitched, the IS will be conceptually wrong. This can also be seen in the simulation study in the next section. Finally, consider the case when a perfectly correlated sequence (such as in a very large composite trade) is interspersed with small metaorders of opposite side (buy/sell), as depicted in Exhibit 6, a situation that often arises due to a volatility adjustment or short-term alpha. This type of pattern would drastically reduce the IS effect as measured by stitching together. In the case of Exhibit 6, the EFFS formula is more robust because the metaorder of small size yields a correspondingly small EFFS contribution.

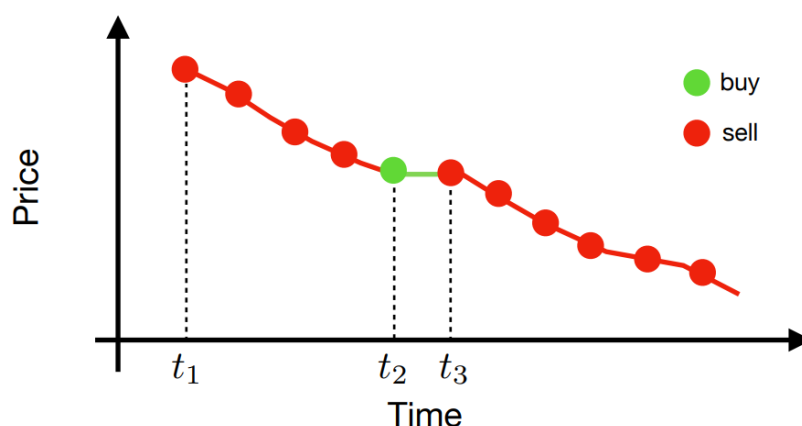


Exhibit 6. A stitched sequence of 10 sell metaorders interrupted by a buy metaorder. As in the previous exhibit, each green/red circle represents a buy/sell metaorder and the times t_1 , t_2 and t_3 denote the commencement of each composite order. The green/red lines connecting buy/sell metaorders represent composite *stitched* metaorders. Without the (small) buy order in the middle, the stitching approach would treat the entire sequence as one composite metaorder benchmarked to the decision price at t_1 . However, with the buy order it is treated as two composite sell metaorders benchmarked at t_1 and t_3 (as well as the buy metaorder at t_2). This has a large effect on the implementation shortfall measure, even though we would expect that a very small buy order would have a negligible effect on the overall impact of the sequence of (larger) sell orders.

A simple way one might think about the above is the following. In situations where the future flow is known, composite metaorder IS seems like an adequate measure. When future flow is unknown, it is impossible to measure the composite metaorder IS, and that is precisely where the EFFS measure comes in. However, this intuition is not general enough. For example, if we were to know the future flow in Exhibit 6, composite metaorders would still be a poor measure, unless we as a human were to look at it and say, rather than three composite metaorders, it

should be treated as one given that the middle order is negligible. But this is a discretionary decision that goes beyond simple stitching. The EFFS is the systematic way to do this and to automate this process.

Propagator models

One approach to deal with correlated order flow is the propagator model (Bouchaud et al. 2004). This model makes the simplifying assumption that each metaorder can be treated separately, and that the effects of each metaorder can be summed linearly. For an individual metaorder of size Q_i starting at time t_i^{start} and finishing by the time t_i^{end} , the market impact after the execution has completed is modelled as follows:

$$MarketImpact(t) = H(Q_i, \sigma_D, V_D) \cdot G(t - t_i^{end}).$$

Here the function H describes the price impact at the end of the metaorder which is commonly modelled by the square-root law as being proportional to $\sigma_D \sqrt{|Q_i|/V_D}$, where σ_D and V_D are the price volatility and the traded market volume during the time horizon of the execution. Furthermore, the (typically) monotonically decreasing function G describes the decay this impact experiences through time, a phenomenon sometimes referred to as *relaxation*.⁴

Using the propagator model, the market impact of each realized trade can be subtracted from the observed prices to obtain *cleaned prices*, i.e., the prices that would have been observed without the impact of the trading. These hypothetical prices can be used to calculate the cleaned P&L of the strategy. The difference between this cleaned P&L and the realized P&L provides a measure of the additional effect on P&L beyond slippage, similar in spirit to the quantity captured by the EFFS. It is useful to contrast the EFFS formula and the propagator model in the context of the implementation shortfall formula: the EFFS formula reflects both the execution cost and the opportunity cost part of the implementation shortfall, while the propagator model only reflects the execution costs.

Despite the conceptual simplicity of the propagator model, there are several hurdles to overcome to fit it. Estimation requires access to market volumes, which may not be available or hard to access for some OTC markets, especially historically. It also typically requires a long history of trading data, with potentially millions of metaorders required for adequate estimation. This provides the propagator model limited adaptivity, thus making it ill-suited to identifying changes in execution quality. Furthermore, it requires the fitting of multiple linear regressions which can be challenging in practice due to the number of inputs, collinearity, hyperparameter choices, etc. Indeed, there are even choices to be made on basic input variables, for example whether to use calendar time, sample time, or volume time. We present a numerical comparison of the propagator model and the EFFS model in the next section.

Consider the EFFS formula:

$$EFF \text{ Shortfall (EFFS)} = (A_{i+1} - A_i) \ E_i \left[\sum_{j=1}^N Q_{i+j} \right].$$

The formula consists of two parts: (i) the difference of the decision prices, $A_{i+1} - A_i$, and (ii) the EFF, i.e., the expected future metaorder flow $E_i[\sum_{j=1}^N Q_{i+j}]$. Recall that the expected sum of future metaorders is truncated after N terms. Intuitively, N is the number of periods/days such that the correlation of signed metaorders separated by this number of periods is

⁴ Typical examples are power-law decay $G(t) = t^{-\beta}$, $\beta > 0$ and exponential decay $G(t) = e^{-\kappa t}$, $\kappa > 0$.

negligible, so the market impact of metaorders more than N periods later is close to zero in expectation. This is a fixed hyperparameter and later we will show how to set it.

Computing the difference in decision prices, $(A_{i+1} - A_i)$, is usually done ex post. Note that this still allows for real-time updates, where the EFFS is evaluated at the end of each period/day. Given the level of noise typically present in trade data, one usually averages EFFS over periods of at least a month. Thus, a one-day lag due to the ex post calculation has little influence on the rolling estimate.

The bigger challenge is to estimate the EFF, given by $E_i[\sum_{j=1}^N Q_{i+j}]$, which is the subject of this section. The EFF might be deduced from the specific trading strategy or it might need to be estimated from historical data. Below, we consider both cases.

Once we have estimated both terms, we can simply multiply them to obtain the daily EFF shortfall (this is done ex post as we will explain). The sum of the EFF shortfalls for each metaorder yields the total dollar EFFS.

Implementing the EFFS formula

Computing the EFFS when target position or trade sequences are known in advance

The first step in computing the EFFS, namely obtaining the daily EFF, given by $E_i[\sum_{j=1}^N Q_{i+j}]$, is the most involved. Depending on the investment decision process, it can be approached in different ways.

The simplest case is when the terminal target position (after many metaorders) is known, potentially as an output of the specific trading system. The expected future flow is:

$$E_i \left[\sum_{j=1}^N Q_{i+j} \right] \approx TargetPosition_i - CurrentPosition_i - Q_i.$$

Thus, in this particular case, there is no need to estimate the expectation on the left-hand side of the equation above, and therefore no requirement to choose a value for N . If we know the target position, why is this an approximation? It is possible due to trading conditions and constraints that the target is not met. For example, suppose our goal is to get to a target position of X from 0 over a period of 10 days. However, there is a constraint that the system cannot trade more than 10% of average daily volume. It is possible that it will take more than 10 days if liquidity dries up. Furthermore, we assume that the effect of the initial order Q_i on any order flow after reaching the target position is approximately zero. However, this is a very special case. In practice, we usually do not know the target position and need to resort to the approaches below.

In some cases, we not only know the target position, but also the details on how to trade into this position. Most commonly, this is the case when the alpha/signal construction process is transparent, i.e., not part of an opaque machine learning algorithm, and available to the execution agent. In this case, the expected future flow can be deduced directly. A simple example could be an event-based signal that would change its value from 0 to 30 lots in steps of 10 lots on three consecutive days/metaorders after the triggering event (1 lot being 100 shares). The expected future flow (after the event and the initial 10 lot trade) would go down from 20 lots to 0 in steps of 10 lots.

Table 1 provides the details of this event-based signal example. Note that the desired position and the resulting metaorders are determined by the strategy and we know this in advance. The decision prices are the ones observed in the market, e.g., each day's opening price. Given this information (columns 1-4), we now compute the EFFS. First, given that the sequence of metaorders is known in advance, we can infer the EFF at each day by simply summing up future metaorders, as shown in the fifth column. Next, we compute the price delta, $A_{i+1} - A_i$, in the sixth column. Note that it is the delta between the next and this day's decision price. This is only available ex post, i.e., we compute the price delta for a given day when observing the next day's opening. Now we can obtain the EFFS contribution on each day (seventh column) by simply multiplying the EFF (column 5) and the price delta (column 6). As it depends on the price delta, it is also only available ex post. The total EFFS for a given period is the sum over the daily EFFS for all days in this period. For the entire sample period in the example, the EFFS is thus 30 (in units of \$100).

Consider the mechanics of Table 1. The trade is triggered by an event and we know the target position is 30 and we aim to trade over three days. Again, we are abstracting from any impact beyond three days and assuming that we are unconstrained and able to do trades of 10 per day. The initial decision price is 101 and the trader begins with a metaorder of 10. The initial trade and/or general market conditions moves the price up by 2. The trader knows there are 20 more to go, hence the EFFS is $2 \times 20 = 40$ (given that the price moved by 2 prior to executing those). The next day the trader executes 10 lots but the price drops by 1. The trader has 10 lots to go and the EFFS is $-1 \times 10 = -10$. The third day the trader executes the last leg, another 10 lots, and the price goes up by 3. The total sum of EFFS is $40 - 10 = 30$. Let us for simplicity assume that the three trades filled at 103, 102, and 105.

Time [Day]	Desired position [lots]	Metaorder size [lots]	Decision Price [\$]	EFF [lots]	Price delta [\$]	EFFS (shortfall) [\$100]
1	0	10	101	20	2	40
2	10	10	103	10	-1	-10
3	20	10	102	0	3	0
4	30	0	105	0	1	0
5	30	0	106	0	-	-

Table 1. Example illustrating the computation of the EFFS for a strategy where a transparent investment process allows us to infer future flow in advance: we know on day one that we are entering a position of 30 lots over the course of the next three days. Here the EFF can be obtained without estimation (given our assumptions of zero impact on future order flow after three days).

The standard IS applied to each metaorder yields $+10 \times 2 (=103-101) - 10 \times 1 (=102-103) + 10 \times 3 (=105-102)$ which is 40. However, benchmarking to the initial decision price, in other words viewing it as one composite metaorder, it is $+10 \times 2 (=103-101) + 10 \times 1 (=102-101) + 10 \times 4 (=105-101) = 70$. The difference is the EFFS. The EFFS measures the hidden impact cost which is 30 ($=70-40$). Note that benchmarking to the initial arrival price is identical to the stitching of metaorders into composite metaorders. While this approach produces an identical answer to using the EFFS in this simple setting, it is flawed in general, as we highlight in the discussion surrounding Exhibits 5 and 6 in the previous section.

This approach may be very effective and avoids estimation errors given that the EFF can be obtained exactly from the strategy. However, it may be challenging if the set of signals is large,

diverse and involves many different signal smoothing techniques, and may be infeasible if the signal construction is not transparent.

Data-driven estimation of the EFF shortfall

The next approach is data-driven and is arguably the most versatile, being applicable to any strategy. We use this approach for the empirical study of actual trade data.

In general terms, the idea is to use a statistical model, such as a parametric model involving some predictors, to infer the EFF given by $E_i[\sum_{j=1}^N Q_{i+j}]$. Once the model is fit to a set of historical training data, we can then make live predictions during runtime. We find that metaorder size, Q_i , is often enough to infer the EFF with reasonable accuracy. We thus use Q_i as our only predictor. Intuitively, this makes sense: large size metaorders are likely to be followed by large metaorders of the same side (buy or sell), while small metaorders are likely to be followed by small metaorders with lower probability of consistency of the side (so the EFF is closer to zero).⁵ Given the predictor variable, we now have to formulate the statistical model.

In practice, we have found that a piecewise constant function works well. To do so, we split the range of possible values of Q_i in b intervals or buckets.⁶ The prediction for each interval is simply the historical mean over the historical future flows, $\sum_{j=1}^N Q_{k+j}$, of each order Q_k in that bucket. Mathematically, we can write this as:

$$E_i \left[\sum_{j=1}^N Q_{i+j} \right] \approx \text{Mean} \left[\left\{ \sum_{j=1}^N Q_{k+j} ; \text{ for all } k, \text{ such that } Q_k \text{ is in same bucket as } Q_i \right\} \right].$$

We make this model more statistically robust by exploiting the following symmetry: If a buy metaorder of size Q has an EFF of a given value x , we expect the equivalent sell metaorder of size $-Q$ to have an EFF of value $-x$. We can then group the metaorders in intervals of absolute size $|Q_i|$. To obtain an estimate of the future flow of a given order Q_i , we first identify the size bucket in which Q_i resides and denote this by b^* . Then, restricting attention to those orders Q_k in bucket b^* , we compute the mean of the “side-adjusted” future flows $\text{sign}(Q_k) \sum_{j=1}^N Q_{k+j}$. Lastly, we then restore the side, multiplying by $\text{sign}(Q_i)$. This can formally be written as:

$$E_i \left[\sum_{j=1}^N Q_{i+j} \right] \approx \text{sign}(Q_i) \text{Mean} \left[\left\{ \text{sign}(Q_k) \sum_{j=1}^N Q_{k+j} ; \text{ for all } k \text{ such that } |Q_k| \text{ in same bucket as } |Q_i| \right\} \right].$$

This is the approach that we use in our empirical study in the next section. To clarify and explain each step in more detail, we also present an example of this implementation in Appendix C.

⁵ This is related to long-range autocorrelations in trade flow.

⁶ There are different methods to partition the data in buckets. Here we use a quantile-based approach where the boundary of the intervals/buckets are given by quantiles of the distribution of Q_i . For example, to obtain four buckets we would split at the 25th, 50th and 75th quantile.

Simulation and Empirical Results

We now show that the EFFS formula gives estimates that are consistent with the challenging to implement propagator models of market impact and it avoids the flaws of the stitching approach. First, we report the results of a simulation study with market dynamics driven by a propagator model of market impact. Second, we report results of applying the EFFS to a set of proprietary order flow and execution data from Man Group. We then compare the EFFS and the propagator model fitted to the same data.

Results from a simulation study

In the simulation study, we make the simplifying assumption that the trades contain no alpha, which we believe is a reasonable approximation in practice when the holding period is long with respect to the execution horizon. That is, if the trade is designed to capture alpha that will be realized over one month and the execution happens in one day, it is safe to assume that the alpha will not all be realized in the execution day. We also assume that there are no spread costs and the trades can be completed at mid-prices. These simplifying assumptions ensure our focus is on market impact effects across the sequence of metaorders.

We consider 10 periods of trading – which can be thought of as days during which each metaorder is executed, except the final special point in time, which will be used to mark-to-market our position. Think of the final point in time being well after the execution has been completed. The price (before allowing for impact effects) is a simple continuous random walk, with normal innovations between the start of each period and total volatility of 100 bps during each of the 10 periods.

Using a simple propagator model for the data generating process, each trade is assumed to add a $0.1\sigma\sqrt{|Q|}$ increment to the arrival price in the direction of the trade. This is equivalent to the immediate impact function described earlier which follows the square-root law of market impact. This initial impact is assumed to decay partially, so that by the next arrival time the individual trade effect is reduced⁷ to $0.05\sigma\sqrt{|Q|}$, and then to $0.02\sigma\sqrt{|Q|}$. All trade impact reverts to zero (i.e., completely dissipates) by the final period. The final period is special – it can be thought of as a longer gap like year-end. In practice, marking to market the P&L at observed prices without sufficient time passing for the impact to revert may lead to overly optimistic P&L estimates. For simplicity, in this simulation, there is no permanent impact. We use a (discrete kernel) function G to represent the above decay formulation rather than a more complex power-law decay. Together, this immediate impact and decay/relaxation specification parallels the stylized behaviour of the propagator model.

We consider illustrative strategies of correlated trades with autocorrelation coefficient of 0.85. Such autocorrelation could be caused by smoothing a raw trading signal. Indeed, an exponentially weighted moving average of a raw trading signal with a half-life of one week would result in an autocorrelation of comparable size. Furthermore, we consider three different cases for the trade innovation, i.e., the surprise or independent component of each trade. We can decompose every new metaorder into a sum of the expected component and the residual that is not predictable. The cases are:

- a) Neutral.** The trade innovations are normal and uncorrelated with past price moves.
- b) Reversion strategy.** The correlation of the normal trade innovation (the part not predictable from past trades) to the most recent price return is -0.2.

⁷ In other words, $G(1) = 0.5, G(2) = 0.2, G(i) = 0$ for $i > 2$.

c) Momentum strategy. The correlation of the trade innovation to the most recent price return is 0.2

Note that the overall correlation between trades and the most recent price returns would be a relatively modest 10% (in absolute value) for the latter two cases.⁸ In general, the correlation is an outcome of the holding period of the strategy and how frequently metaorders are issued. In practice, this correlation can be higher or lower than the value chosen for our illustrations. In our Monte Carlo analysis, we generate 100,000 scenarios, each containing 10 periods as described above. (We can think of the 100,000 scenarios as 100 replications for each stock of a universe of 1,000 stocks, which corresponds to a realistic amount of data.)

The effect we are estimating is the expected effect on P&L by the end period. We do so by using the EFFS formula and two alternative methods: 1) benchmarking all the metaorders to the first arrival price (“First Arrival” in the table below), and 2) metaorder stitching (“Stitching”), as described in the previous section. We report the dollar costs divided by notional traded in basis points in Table 2. In this simple setting we can use the first implementation of the EFFS formula, where the target position is known. Table 2 shows that the EFFS formula adequately captures the P&L effect of trading. Benchmarking to the First Arrival price works only for the Neutral strategy case (as the expectation of First Arrival price is the same as the expected value of the final period price). The First Arrival method does poorly with the reversion and momentum strategies. The Stitching method gives biased results in all three cases. Notably, for the reversion strategy, both the First Arrival and the Stitching method estimate the wrong sign, i.e., suggesting that the trading impact would be to the benefit of P&L. The wrong sign appears because in the reversion strategy we are on average buying in falling markets and selling in rising markets, as also illustrated in Exhibit 5.

	True effect	EFFS model	First Arrival	Stitching
Neutral	-14.07 ± 0.3	-14.14 ± 0.2	-14.05 ± 0.3	-16.29 ± 0.3
Reversion	-13.94 ± 0.3	-14.16 ± 0.2	23.10 ± 0.3	14.16 ± 0.3
Momentum	-13.97 ± 0.3	-14.31 ± 0.2	-51.49 ± 0.3	-47.27 ± 0.3

Table 2. Estimated effects on P&L (in basis points) due to long range price impact. The results are obtained using 100,000 Monte Carlo simulations of 10 trading periods for the three strategies described above. We compute average effect and present ± two standard deviations from the Monte Carlo simulations. Given that we know the underlying impact model, we can compute the true effect. Here we use the first estimation procedure explained in the previous section. Evaluation of the First Arrival and Stitching method are straightforward, but neither yields consistent results.

Results from execution data

In practice, the laws governing market impact and market impact decay are not readily observable and, unlike in the simulation study above, have to be estimated from real trade data. A common approach is to fit a propagator model. Once the market impact model is available, the P&L effect can be estimated by evaluating the fitted propagator model on the realized trades. Fitting a propagator model can be a challenging task for several reasons. The first challenge is data availability (the estimates in the literature use millions of metaorders Brokmann et al. 2015; Bucci et al. 2018). Second, this is a high dimensional estimation problem that is confounded by a potentially significant degree of correlation in regressors. Finally, there are many choices to make, regarding ways of pooling the data, choosing priors, regularization methods and lookback periods. The EFFS method, in contrast, is considerably easier to implement and is much less parametric.

⁸ Given the decomposition of a metaorder as $Q = Q_{expected} + Q_{innovation}$, for the momentum strategy we have $corr(Q_{innovation}, Return) = 0.2$, but $corr(Q, Return) \approx 0.1$ (and similarly for the reversion strategy).

To illustrate the strengths of our method and relative performance to a propagator model approach, we report results from applying both methods to metaorders from a proprietary trading strategy. We report estimates for every month in the period January 2018 to January 2019 in Exhibit 7. To focus on relative performance, we normalize the measurements so that the propagator estimate equals 100% every month. To be clear, a value of 80% means that the EFFS has 80% of the propagator shortfall.

There are two important takeaways from Exhibit 7. First, over the entire sample period, the simple EFFS model performs similarly to the difficult to estimate propagator model. The average difference in dollar terms between the two models over the period is approximately 12% and only 6% if the tumultuous month February is excluded. The performance of the EFFS in February points to the second important observation: the EFFS estimate is more much adaptive as it uses only one month of data. The propagator model, in contrast, uses approximately one year of data, so is naturally smoother and much less sensitive to changes in the distribution of price moves and trades.

The adaptability of the EFFS method is also illustrated by computing the median of the absolute change in shortfall month on month, which is 47% for the EFFS model and only 15% for the propagator model. The relatively high variance in the EFFS estimates is exactly what we expect given the estimates rely on standard execution slippage measurements calculated per metaorder which, as already noted, are extremely variable being strongly affected by price moves during the metaorder.

One might wonder whether it was possible to make the propagator model more adaptive by using say only one month of data. However, as we have emphasized, the propagator model is much more complex and it is often not feasible in practice to fit the model on small samples of data. While each approach has its merits, the EFFS model is much more reactive during periods of sudden change, such as crises.

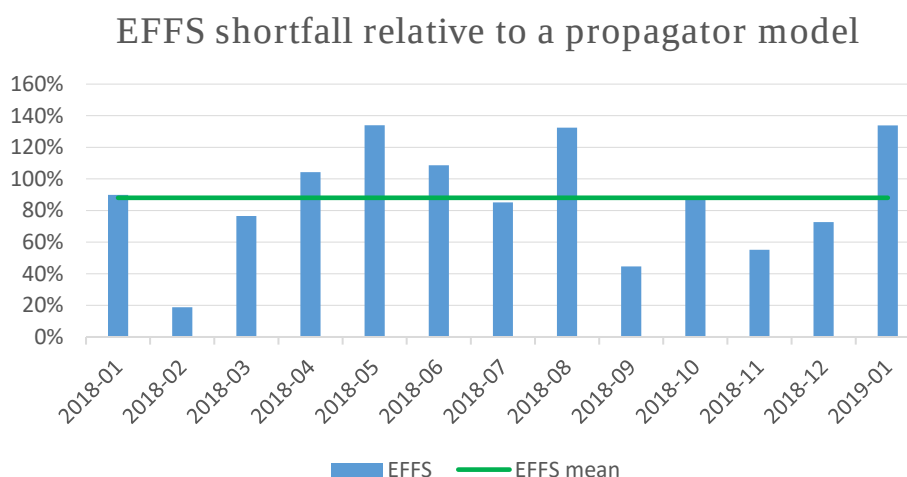


Exhibit 7. For the period from January 2018 to January 2019, we show the monthly average shortfall computed using the propagator model and the EFFS model. Statistics are based on approximately 730,000 trades. The propagator model is normalized to 100% (only relative performance can be shared due to the proprietary nature of the data). Hence, a value of 80% means that the EFFS model returns a shortfall that is 80% of the size of the propagator model. The EFFS model gives estimates that are close to estimates from the propagator model when averaged over the entire year. Note the larger difference in the volatile month of February 2018.

Conclusions

In this era of machine learning and big data, trading occurs much more frequently. As such, it is increasingly important to have the most efficient execution of trades. While it is commonplace to divide larger metaorders into smaller child orders to reduce market impact, we investigate the case where metaorders themselves are correlated. Here we introduce a new technique which we call the EFFT, the expected future flow shortfall.

We provide simulation evidence in a number of different market scenarios that shows that the EFFT approach has distinct advantages over other approaches. For example, it is tempting to simply stitch all the metaorders together. However, our evidence shows that this method fails and gives counterintuitive results in periods of price reversion.

We also provide a real trading example using proprietary data. Here we compare the performance of the simple EFFT model and a propagator model. The propagator model is highly parameterized, difficult to estimate, and requires a very large amount of data to calibrate adequately. In contrast, the EFFT model requires significantly less data which allows it to be much more reactive to market conditions.

Our empirical analysis reveals that over the entire sample the EFFT model and the complex propagator model perform similarly. However, there are sharp differences when the results are examined month by month. The EFFT is much more responsive to changing market conditions. Our results show that the EFFT is more than three times as reactive when computing the median absolute change in shortfall month on month.

Being able to evaluate the EFFT and the hidden slippage of a trading strategy on a monthly basis can be important when using it to construct portfolios of strategies which are optimized on a regular basis. Let us further illustrate this use case: Imagine we want to allocate to two strategies, A and B, both with an expected annual performance of 12% before costs and both equal volatility and tail risk. Both strategies have visible costs equivalent to 3%, leading to an apparent after costs performance of 9%. Given the above information a portfolio manager would be allocating equally to these strategies. However, using the EFFT one can evaluate the hidden slippage which could be 2% for strategy A and 6% for strategy B, yielding a true performance of 7% and 3% respectively. Being able to evaluate the hidden costs using the EFFT can lead to significant improvements in portfolio allocation. Not only is the EFFT useful in portfolio construction, but also in alpha research itself. Many data-driven alphas are calibrated to maximize risk-adjusted after-cost performance. It is crucial to take the all costs into account.

In summary, the theoretical properties and empirical evidence point to the benefits of deploying the EFFT formula as a simple way to estimate the trading effects on P&L beyond standard slippage estimates. This is particularly important for faster systematic strategies and those with autocorrelated order flow. Our model provides an alternative measure to both the stitched metaorder and propagator approaches. In particular, our model performs demonstrably better than the heuristic stitched metaorder approach. It also obtains results comparable to the propagator model, while being simpler and less data hungry, allowing it to be more adaptive to changing market conditions.

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Appendix A: An approximation to the EFF shortfall formula

A quick estimate of the order of magnitude of EFFS can be obtained using just a few quantities that are easily empirically measured: the probability p that the next trade is the same sign as the previous one, the notionally weighted average price move to next arrival time, $\overline{Q \cdot (A_2 - A_1)}$, and the average of the signed and absolute trade size, $\bar{Q}$ and $|\bar{Q}|$ respectively. Here the overline denotes the sample average which will serve as our expectations. The contribution to the P&L of future order flow in this case is simply:

$$EFF \text{ shortfall} \approx \frac{\overline{Q \cdot (A_2 - A_1)}}{\bar{Q}} \cdot |\bar{Q}| \frac{2p - 1}{2 - 2p}.$$

This expression would be precise for the case of a Markov chain trade generation process with the trade signs switching randomly at each step with probability $(1 - p)$, and sizes independent of the signs. Indeed, in this case, the expected net order flow in the next step is equal to $(p - (1 - p))E|Q|$; in the next one it is $(p - (1 - p))^2 E|Q|$, and in the $(k + 1)$ -th step, $(p - (1 - p))^k E|Q|$. Summing up the steps to infinity, we get a geometric progression with a value of $\left(\frac{2p-1}{2(1-p)}\right) E|Q|$.

In our simulation study, we estimate $p \approx 0.82$, hence the future flow multiplier is approximately 1.83. With the average price move of 5.65 bps, we get an estimate of the effect (in basis points) on P&L (equal to minus EFFS) of approximately -10 bps, for all three strategies considered. As the conditions of the simplified formula are not met precisely, the estimate is slightly biased, yielding an EFFS of 10 bps (so effect on P&L of -10) compared to the true -14 bps. However, it gives a reasonable estimate and it is much better than stitching or benchmarking to the first arrival price for the Reversion and Momentum strategy when comparing to the estimates in Table 2.

In practice, the above approximation provides a quick estimate, which, if material, can be further investigated by using the full EFFS model via the implementation outlined in the main text.

Appendix B: Benchmarking to many decision prices

If we apply the EFFS formula to a set of consecutive metaorders, we can show that it is equivalent to splitting each metaorder into many new metaorders (of potentially different sides), with each being benchmarked to one past decision price. Let us illustrate this for two metaorders. We write the expressions for the “full cost”, i.e., a sum of (standard) slippage and EFFS, associated with each metaorder’s period:

$$Cost(1) = Slippage(1) + EFFS(1) = (P_1 - A_1) \cdot Q_1 + (A_2 - A_1) E_1\left(\sum_{i=2}^{\infty} Q_i\right)$$

$$Cost(2) = Slippage(2) + EFFS(2) = (P_2 - A_2) \cdot Q_2 + (A_3 - A_2) E_2\left(\sum_{i=3}^{\infty} Q_i\right)$$

Adding and subtracting $E_1(Q_2)$ to Q_2 in the second expression, summing the two expressions and rearranging the terms yields:

$$Cost(1) + Cost(2) = (P_1 - A_1) \cdot Q_1 + (P_2 - A_1) \cdot E_1(Q_2) + (P_2 - A_2)(Q_2 - E_1(Q_2)) + \dots,$$

where ... refers to terms that do not contain Q_1 and Q_2 . This can be interpreted as benchmarking Q_1 to A_1 , benchmarking $E_1(Q_2)$, the part of Q_2 expected at time 1, to its decision price of A_1 and benchmarking the “surprise” or unexpected part of Q_2 , equal to $Q_2 - E_1(Q_2)$, to its decision price of A_2 . Note this corresponds to the “hybrid cost” of (Bordigoni et al., 2021).

This can be generalized to infinitely many past decision prices, so that we arrive at a backward-looking representation of cost and EFFS.

Backward-looking representation

The full cost associated with trading Q_i is:

$$Cost = (P_i - A_i) \cdot (Q_i - E_{i-1}(Q_i)) + \sum_{j=-\infty}^{i-1} (P_i - A_j) \cdot (E_j(Q_i) - E_{j-1}(Q_{i-1})).$$

This should be contrasted to the representation we had before the re-arrangement, which we can call “forward-looking”.

Forward-looking representation

Indeed, adding slippage to the EFFS formula, as derived in the main text, we obtain the full cost associated with the period i as

$$Cost = (P_i - A_i) \cdot V_1 + (A_2 - A_1) \cdot E_1\left(\sum_{i=2}^{\infty} V_i\right).$$

It emphasizes effect of the price action during period i on the future flow.

Appendix C: An example illustrating the data-driven implementation of the EFFS formula

Let us recall that as inputs we have historical data of sequences of decision prices, A_i , and metaorders, Q_i . Furthermore, we have two hyper-parameters which we must choose: N , the number of future metaorders to compute the EFF given by $\sum_{j=1}^N Q_{i+j}$, and b , the number of buckets. In practical applications, values around $N = 30$ (e.g., 30 days or approximately one month) serve as a good starting point. If one is unsure, one can start with a lower value and increase N , stopping when the final value of the EFFS does not change materially upon increasing N , i.e., it has converged. This means that any given order has diminishing influence on any future orders beyond N steps ahead, thus including those would not change the result which is why we truncate the sum at N , rather than say summing up till the end of the date sample. Regarding b , in practice, using 10 buckets is a good rule of thumb. If enough data are available, one can use cross-validation to determine the optimal hyperparameters.

In the example in Table 3, we choose $N = 1$ and $b = 2$ (two buckets) for simplicity. With $N = 1$, we only look one-step ahead for the future flow estimation. This is less realistic than the typical choice of $N = 30$, but it is straightforward to extend this to any arbitrary N and any number of buckets. Given the historical sequence of metaorders and decision prices for each day (column 2 and 3), we now go through each step to arrive at the EFF shortfall. The first step of the computation is to assign each metaorder to an absolute size bucket and compute its side-adjusted (i.e., multiplied by the metaorder sign) future flow.

Time [day]	Metaorder Size [lots]	Decision Price [\$]	Absolute size bucket [lots]	Side- adjusted future flow [lots]	EFF est. [lots]	Price delta [\$]	EFFS est. [100\$]
1	16	100	[10-20)	8	7	3	21
2	8	103	[0-10)	-1	1.5	2	3
3	-1	105	[0-10)	-1	-1.5	-2	3
4	1	103	[0-10)	-3	1.5	2	3
5	-3	105	[0-10)	11	-1.5	-2	3
6	-11	103	[10-20)	6	-7	-3	21
7	-6	100	[0-10)	-	-1.5	-	-

Table 3. Example illustrating the data-driven approach to compute the EFF shortfall. The steps to arrive at each column are explained in the main text.

We choose two buckets, one for metaorders with absolute size between 0 and 10, and one for order with absolute size between 10 and 20. We now assign each metaorder to its corresponding bucket resulting in the fourth column. Next, we compute the side-adjusted future flow for each metaorder: For each metaorder we simply sum the sizes of the N subsequent orders and multiply this by the sign of the original metaorder. The resulting values for $N = 1$ are shown in the fifth column.⁹

The next step is to compute the EFF estimate as an average over absolute size buckets. The first bucket, [0-10), contains four metaorders for which we computed the side-adjusted future flow (rows 2-5) with values -1, -1, -3, 11. The average is thus

⁹ If N would have been 2 then the first value would have been $8-1=7$

$$\text{side adjusted future flow for bucket } [0,10) = \frac{-1 - 1 - 3 + 11}{4} = 1.5.$$

In the same way we obtain

$$\text{side adjusted future flow for bucket } [10,20) = \frac{8 + 6}{2} = 7.$$

To obtain the estimated EFFS for each metaorder we simply take the average side-adjusted future flow for the corresponding bucket and multiply it by the sign of the original metaorder. The resulting values are shown in the sixth column. The remainder of the computation proceeds as in the earlier example: We take the price (column 3) and compute the price delta (column 7). Next, we multiply the price delta (column 7) with the estimated EFF (column 6) to obtain the EFFS for each metaorder (column 8). Summing over the EFFS for each meta order during the whole period yields the total EFFS estimate for the period, which in this case is 54 (in units of \$100).

The mechanics are as follows. The initial decision price is 100 and the trader put in an order for 16. Given the order size of 16, one can estimate that the future order will be 7 (based on a historical calibration with buckets – but in this case we are using the actual data to do this). Given the metaorder moved the price by 3, we multiple the 7 times 3 to get the EFFS estimate of 21 in the final column of Table 3. Day two arrives and the decision price is 103. The metaorder is 8. Given the order of 8, we try to estimate the future order. The historical calibration for this bucket is 1.5. Given the price moved by 2, the EFFS is 3 (in eighth column). On day 3, the decision price is 105. The metaorder is of size -1. Given the -1 order, you now expect the next trade to be of size -1.5 given we are in the small size bucket and selling. The price had dropped on the first sell by 2, the EFFS is 3. This continues through day 7. The total EFFS is just the sum of each individual day's EFFS.

The above example should allow the reader to quickly implement the EFFS model. Furthermore, we include in the supplementary material a spreadsheet with another example. The spreadsheet allows one to copy one's own historical time-series of metaorder sizes and decision prices, and choose the hyper-parameters N and b , to automatically obtain the series of daily EFFS. Note that given the low signal-to-noise ratio, one generally requires thousands to tens of thousands of observations to obtain robust estimates.